Explainable AI for Academic Integrity Verification

Progress since Meeting 3

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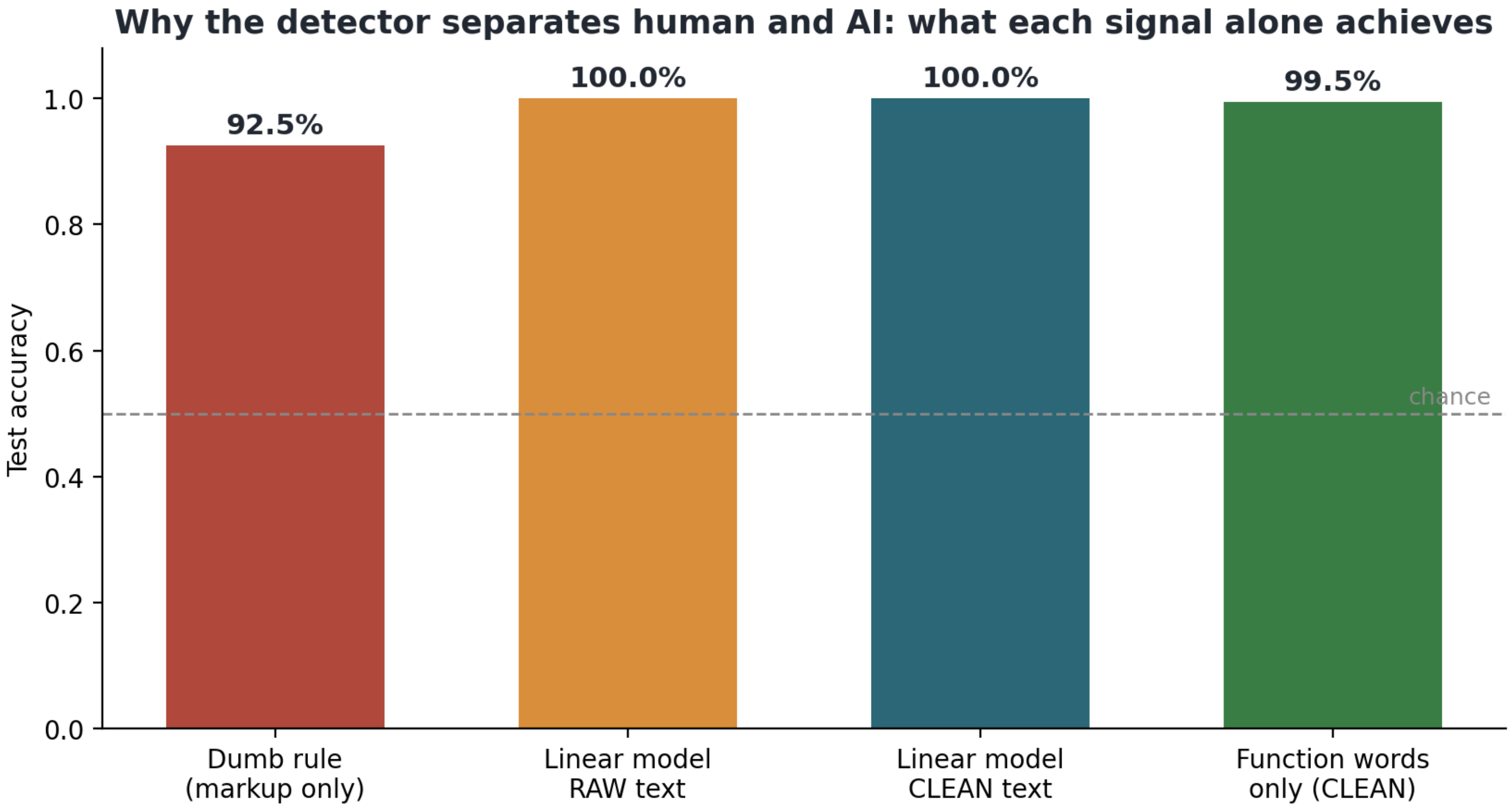
Since Meeting 3 (16 June)

- Where we were: a 1,280-essay dataset and a first detector, audited.
- Explained the detector, which is the whole point of the project (SHAP, Integrated Gradients).
- Tested its limits on an external benchmark (other AI models, other kinds of text).
- Added statistical rigour so I am not over-claiming (confidence intervals, seeds, a locale control).
- Built the first slice of the core contribution: a Verification Interview Guide.
- Stood up the primary, judge-free evaluation and got a result I did not expect.

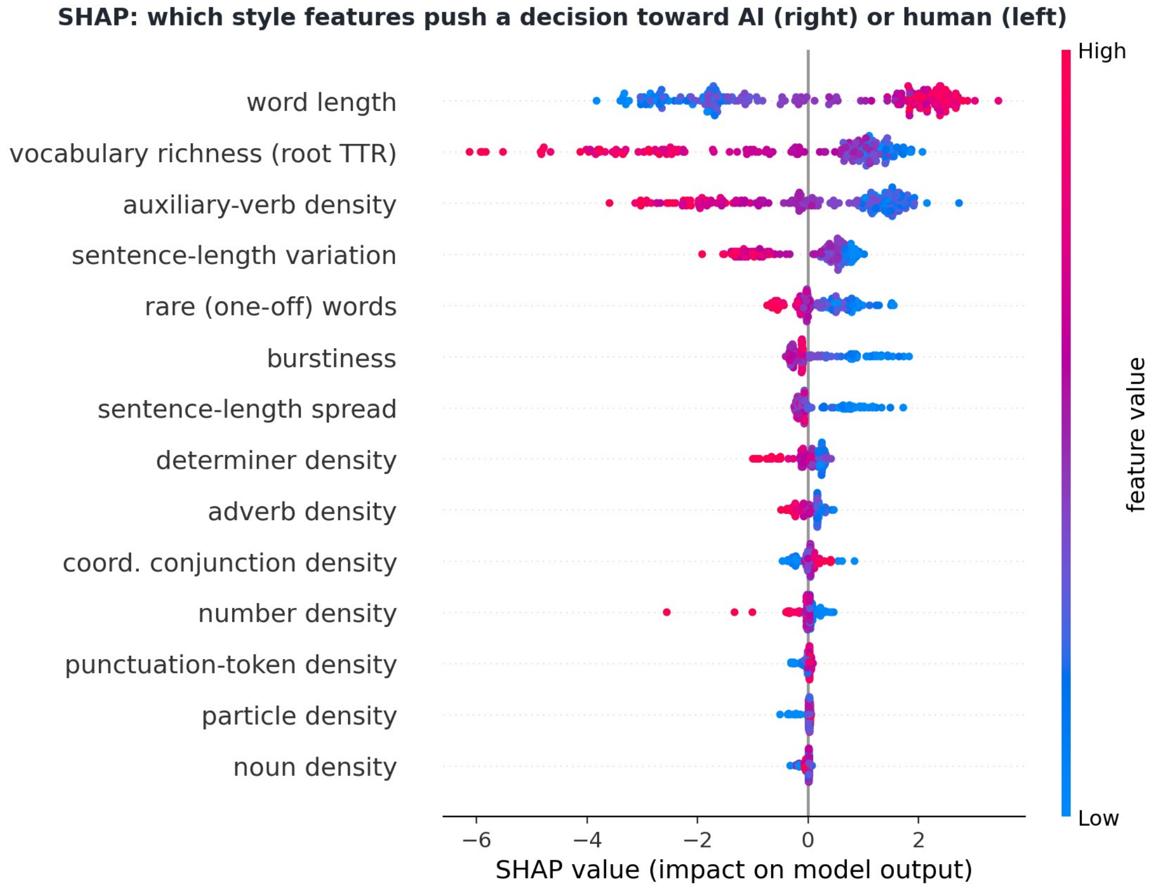
Recap: the detector is honest, not just accurate

The near-perfect score survived an audit. The corpus left formatting tags in the human essays, and a 'has-a-tag' rule alone scored 92.5%. After stripping all formatting from both sides, the classes are still separable using only function words (99.5%), so the real signal is writing style. DeBERTa F1 0.99, RoBERTa 0.995.



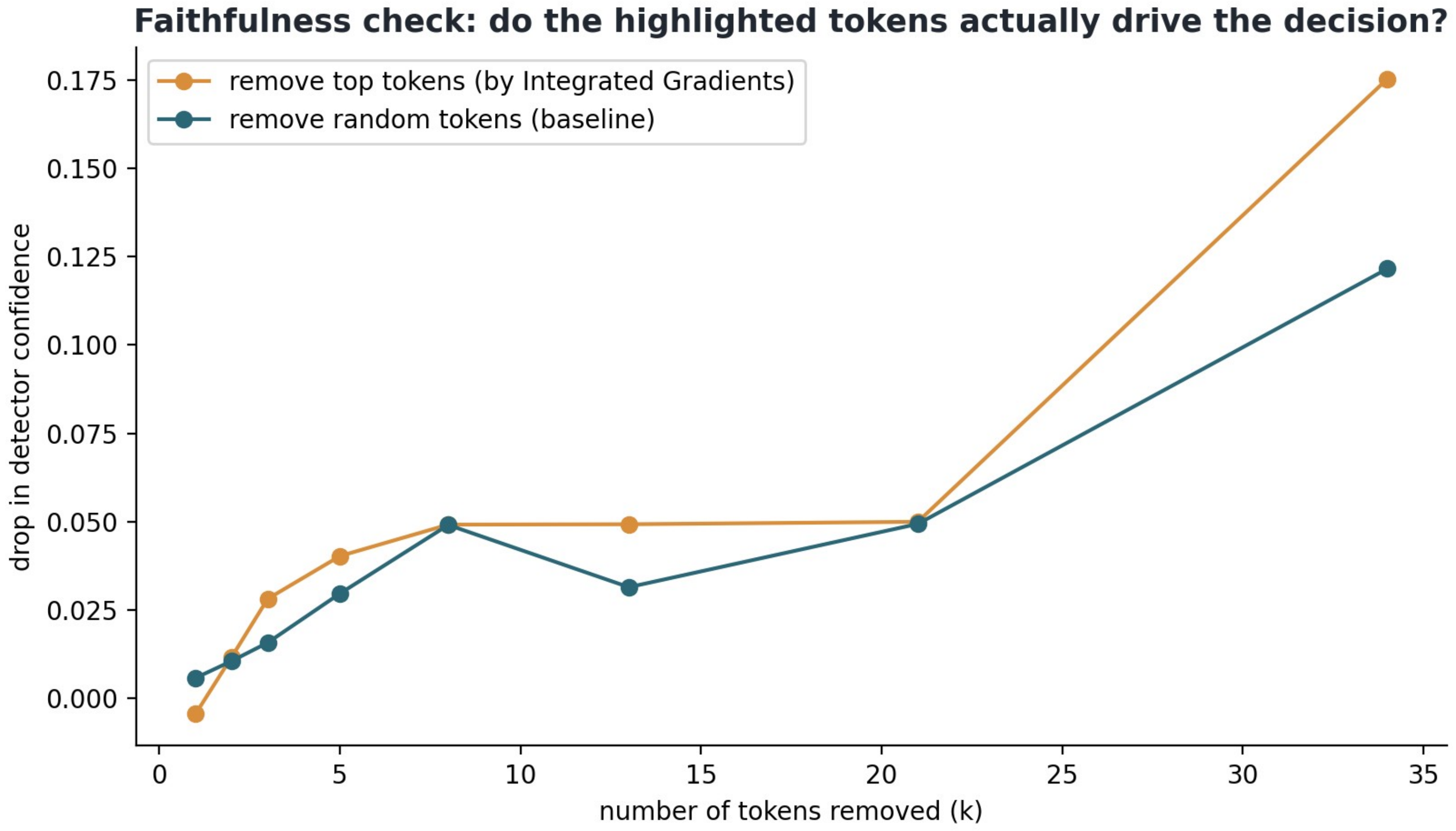
Explaining the decision: the project's whole point

A style-only detector (F1 0.985) gives a lecturer an explanation they can defend. The SHAP plot carries the direction: longer average word length pushes toward AI; richer vocabulary and more varied sentence length push toward human. This is interpretable evidence, not a black-box percentage.



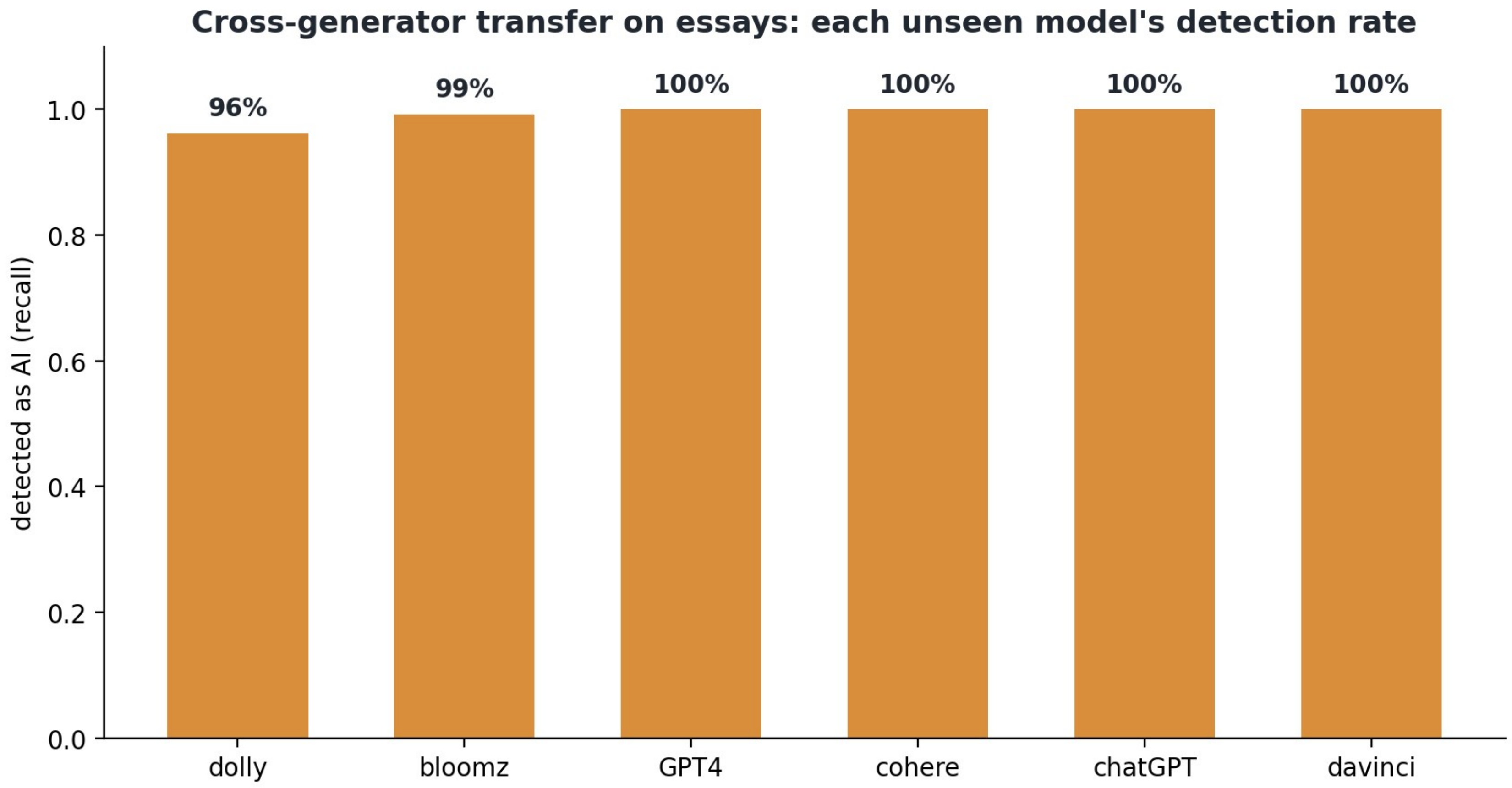
Which explanation can a lecturer trust?

I tested the explanations, not just produced them. Token-level Integrated Gradients on the transformer is diffuse and only weakly faithful: removing the 'most important' words barely changes the decision. The style-based SHAP explanation is the faithful one, so that is what I would put in front of a lecturer.



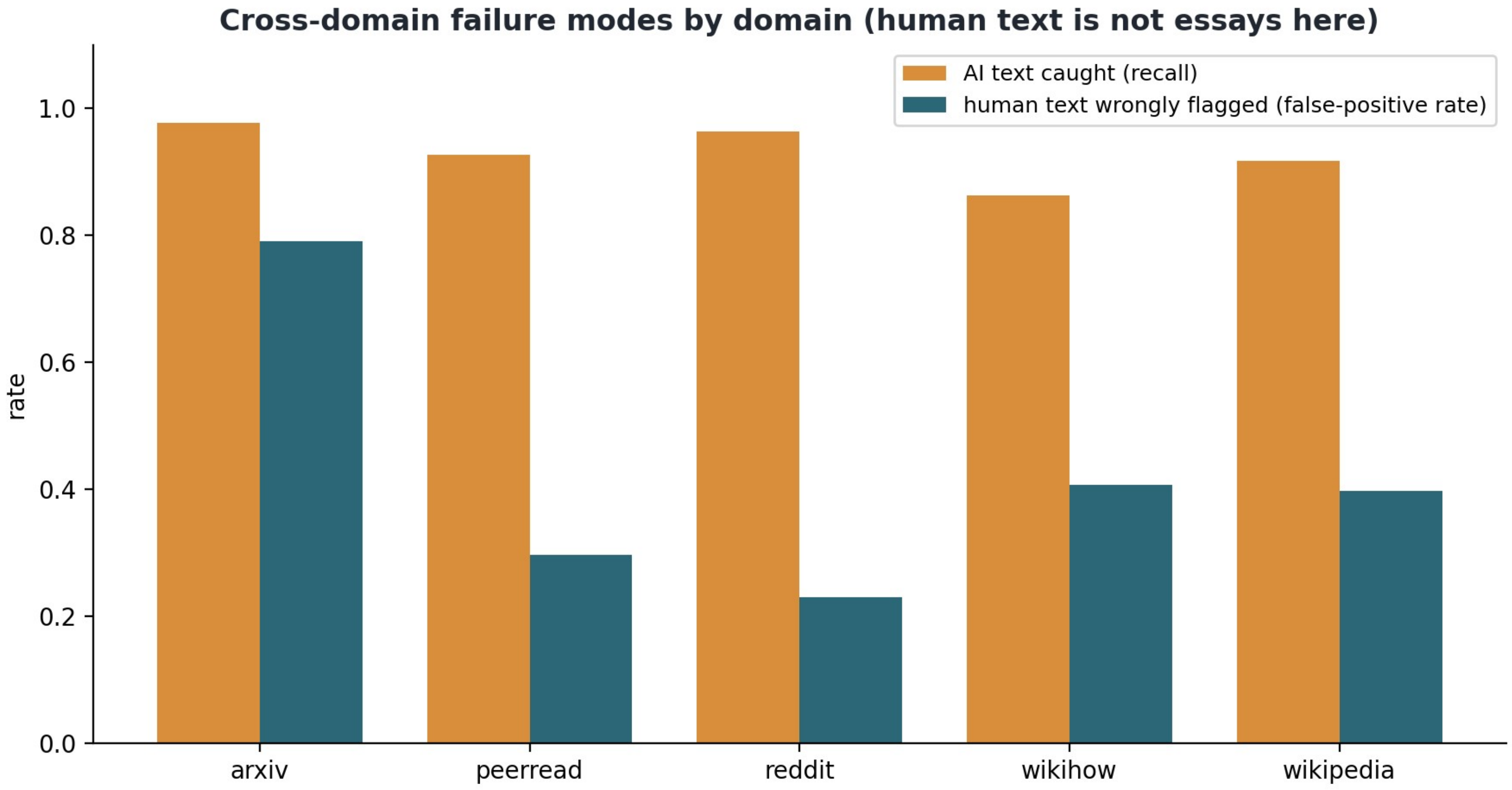
Does it catch AI models it never trained on?

Yes. On the OUTFOX essay split (shipped inside the M4 corpus), against six generators it never saw, cross-generator F1 is 0.97 (n=4,800). GPT-4, ChatGPT, Cohere and davinci are caught essentially 100% of the time; two of the six (dolly, bloomz) sit a little lower. The detector generalises across the model that wrote the text.



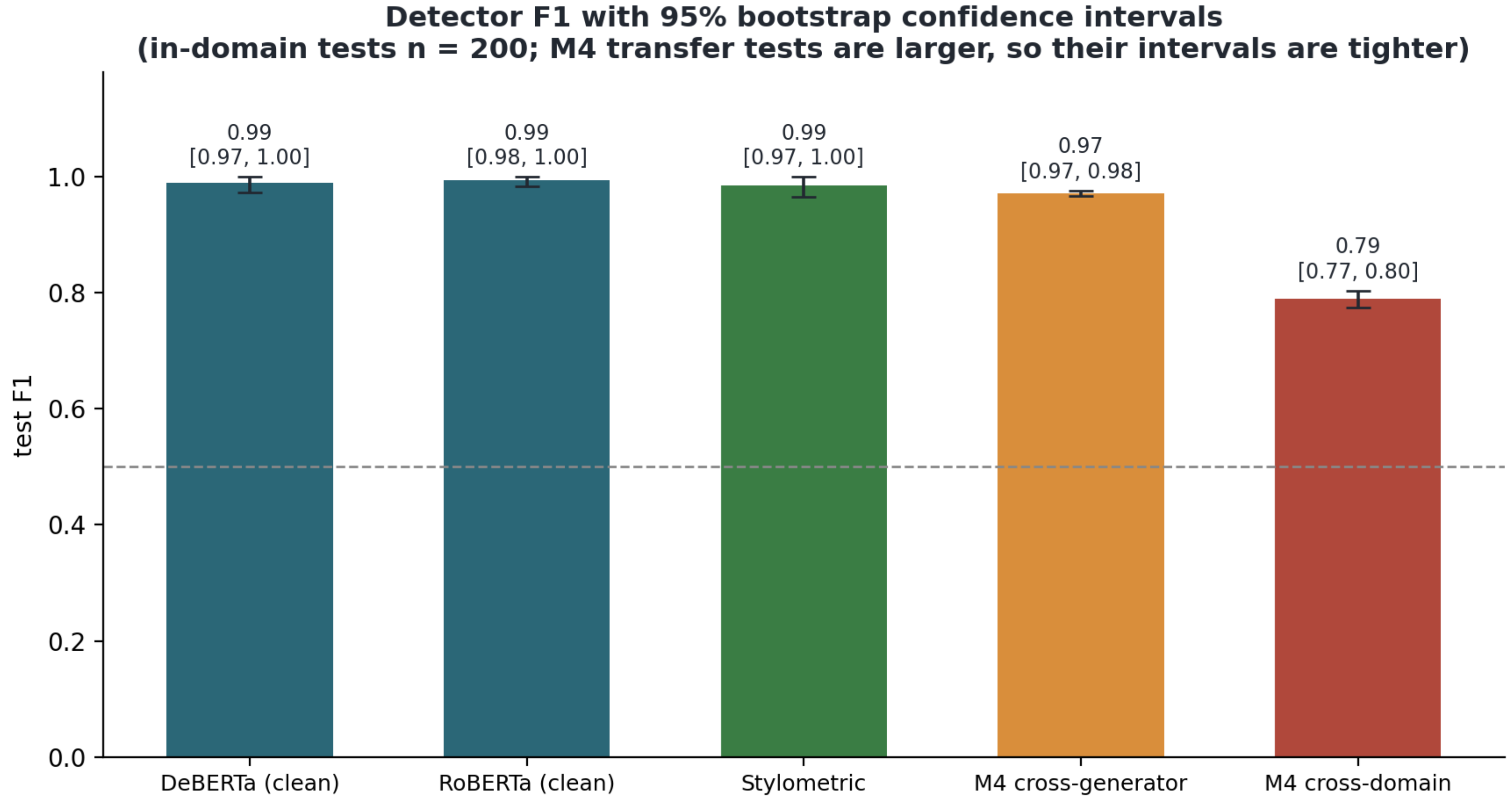
Where it breaks, and why that matters most

On the M4 / SemEval web and academic domains it gets fragile: cross-domain F1 drops to 0.79. The failure mode is the dangerous one, false-accusing humans. Real human arXiv abstracts are flagged as AI 79% of the time. This is the fairness risk the whole project exists to address, now measured rather than assumed.



Rigour, so I am not over-claiming

Bootstrap 95% confidence intervals on the in-domain test: DeBERTa 0.99 [0.97, 1.0] and RoBERTa 0.995 [0.98, 1.0] overlap, so on 200 essays there is no detectable difference between them. Seed-to-seed variation is zero for the style model, and a locale control shows British vs American spelling is not what it relies on.



First slice of the core contribution

A pipeline demonstration. From a flagged essay the system pulls the main claims, cites each to its exact sentences (provenance is guaranteed: the quote is looked up by number, so the model cannot invent one), writes verification questions, and Bloom-tags them. Shown on a sample essay; the next slide evaluates whether the questions actually work.

Claim (cited to sentences 20, 36, 42 of the submission): the author argues that language is a powerful tool that conveys meaning, evokes emotion, and shapes how we understand the world.

Q1. What specific examples from the texts you analyzed did you use to illustrate how language can convey meaning?

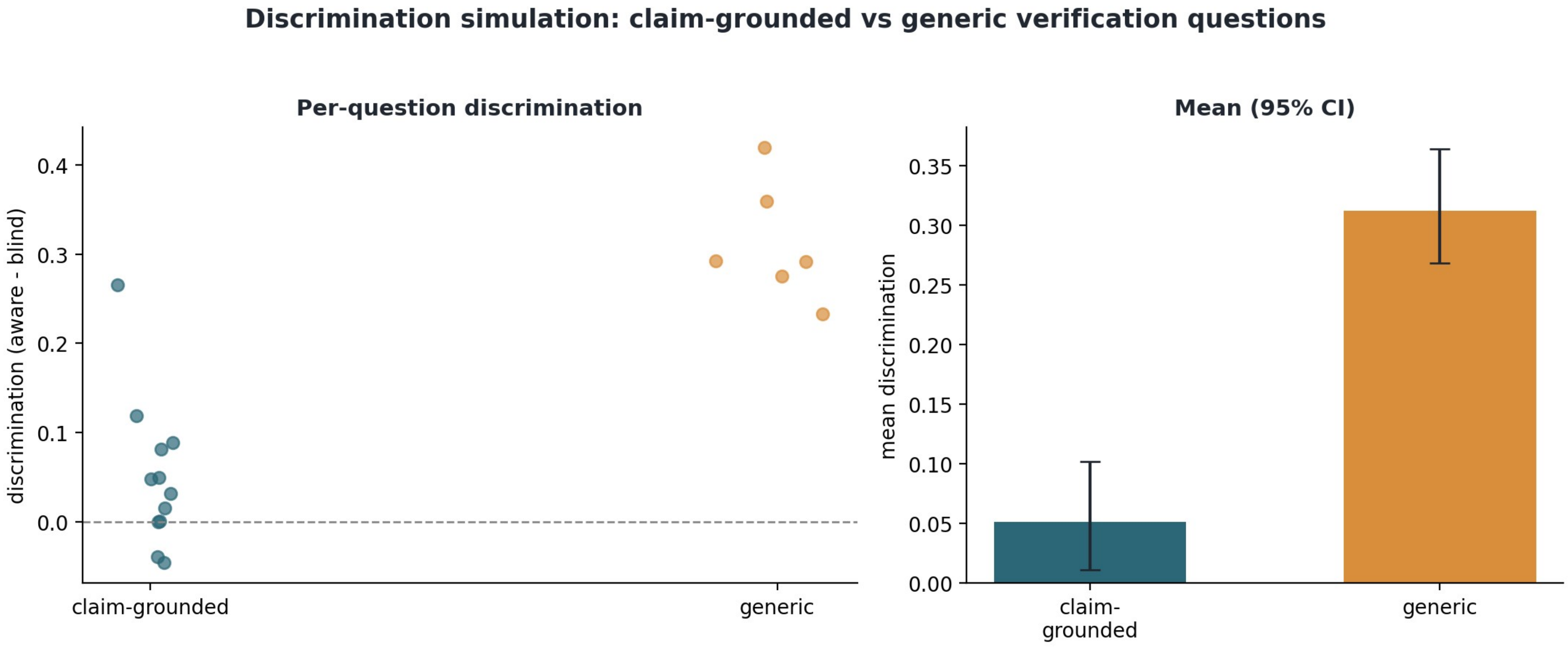
Bloom: understand

Q2. How does the metaphor in one of these texts contribute to evoking a sense of longing and yearning, and what makes this relevant to the human experience?

Bloom: analyse

Evaluating the questions: an early surprise

The primary evaluation needs no human and no LLM judge: a model answers each question with the source and without it, and the gap is the discrimination score. On one flagged essay (18 questions, local model) generic 'explain your own argument' questions discriminate more (0.31) than content-specific ones (0.05). An early, single-essay signal, not a settled result; I will confirm it at scale.



Decisions I need from you

- Backend A (commercial) API key and a small budget: can ATU provide one? It is needed for the core commercial-vs-local comparison.
- Backend B size: I will smoke-test 8B QLoRA on the 4060 this week. If it does not fit, do you pre-approve a 3B fallback (my pick: Qwen2.5 3B)? This changes locked scope.
- Re-scope argument mining to prompted claim extraction: keeps an end-to-end path, but it drops the trained BIO + relation model, leaves the Persuasive Essays datasets unused, and has no quantitative eval yet. Worth it to protect the writing weeks?
- AICS submission: can you forward the call? Irish AICS, Student Track, EasyChair, ~October, and confirm you are happy with the detector-audit / fairness angle.
- Literature review: Chapter 2 is still a skeleton and on the critical path. Which sources do you consider essential?

Plan to submission

- Now to mid-July: question generation (both backends, pending your decisions) + Bloom's classifier.
- Mid-July to early August: integration into the Verification Interview Guide + full evaluation (scale the simulation, add the LLM-judge view).
- August: writing, revision, submission. These weeks are protected; if anything slips I cut analysis depth, not write-up.
- Live deliverables already drafted: a 40-page working draft in the 2026 template (Chapters 1 and 3 to 8 substantive, Chapter 2 still a skeleton), and an AICS Student-Track paper.
- Headline contributions so far: a detector I can explain and defend, a measured fairness risk, and a finding that reshapes how verification questions should be written.